

Group-A (Each question carries 2 marks)

1. Difference between Data and Information with a suitable example.

- * Data: Raw, unorganized facts or figures without inherent meaning (e.g., 25, Ram).
- * Information: Processed, organized data that provides meaning and context (e.g., "Ram is 25 years old"). Information is derived from data.

2. Define DBMS with an example.

* DBMS (Database Management System): A software system that enables users to create, maintain, and control access to a database. It manages data storage, retrieval, security, and integrity.

* Example: MySQL, Oracle.

3. Write down the basic tasks of DBMS.

* Basic DBMS tasks include data definition (schema creation), data manipulation (CRUD operations - Create, Read, Update, Delete), data security and integrity enforcement, concurrency control, and data recovery/backup.

4. Define Metadata with an example.

* Metadata: Data about data. It describes the characteristics and structure of data, such as data type, size, and constraints.

* Example: For a column Age, its metadata would include Data Type: Integer, Length: 3, Constraint: NOT NULL.

5. Define Entity and Attribute.

* Entity: A real-world object or concept about which data is stored (e.g., Student, Book).

* Attribute: A property or characteristic that describes an entity (e.g., for Student, attributes are StudentID, Name).

6. Explain the concept of a schema in DBMS.

* Schema: The overall logical and physical design of a database. It's a blueprint defining the structure of tables, attributes, relationships, and constraints, without containing the actual data.

7. Write down the different types of relationships in DBMS.

* The main types of relationships in DBMS are:

* One-to-One (1:1): Each record linked to exactly one.

* One-to-Many (1:M): One record linked to multiple, but multiples linked to only one.

* Many-to-Many (M:N): Multiple records linked to multiple.

8. List the responsibilities of DBA.

* DBA responsibilities include database installation and configuration, security management (user access), backup and recovery, performance monitoring and tuning, and database maintenance.

9. Define weak and strong entity. Give an example.

- * Strong Entity: An entity that can exist independently and has its own primary key (e.g., Employee).

- * Weak Entity: An entity that depends on another strong entity for its existence and primary key (e.g., Dependent, which needs an Employee).

10. State the differences between strong and weak entity in ER diagrams.

- * Strong Entity: Represented by a single rectangle, has its own primary key.

- * Weak Entity: Represented by a double rectangle, depends on a strong entity for its identification (no independent primary key).

11. Define primary and candidate key with an example.

- * Candidate Key: An attribute or set of attributes that can uniquely identify each row in a table (e.g., StudentID, Email for a Student table).

- * Primary Key: A candidate key chosen to uniquely identify each row; it must be unique and NOT NULL (e.g., StudentID if chosen as the main identifier).

12. Explain the difference between a primary key and unique constraint.

- * Primary Key: Uniquely identifies rows, cannot be NULL, only one per table.

- * Unique Constraint: Ensures uniqueness of values in a column/set of columns, can allow one NULL value, multiple per table.

13. Discuss Data Independence with a suitable example.

- * Data Independence: The ability to modify the database schema at one level without affecting the schema at higher levels.

- * Example (Physical): Changing data storage type (HDD to SSD) should not affect application logic.

- * Example (Logical): Adding a new column to a table should not break existing applications that don't use that column.

14. Difference between generalisation and specialisation. Give an example.

- * Generalization (Bottom-Up): Combining lower-level entities with common attributes into a higher-level entity (e.g., Car, Truck $\rightarrow$ Vehicle).

- * Specialization (Top-Down): Breaking a higher-level entity into lower-level sub-entities based on specific characteristics (e.g., Employee $\rightarrow$ FullTimeEmployee, PartTimeEmployee).

15. Explain the cardinality ratio with example.

- * Cardinality Ratio: Defines the number of instances of one entity associated with instances of another entity in a relationship.

- * Example: A Department has Many Employees (1:N relationship).

16. Define cardinality and degree.

- * Cardinality: The number of tuples (rows) in a relation (table).

- * Degree: The number of attributes (columns) in a relation (table).

17. Define DDL and DML.

- * DDL (Data Definition Language): SQL commands to define and modify database structure (e.g., CREATE TABLE, ALTER TABLE).

- * DML (Data Manipulation Language): SQL commands to manage data within tables (e.g., SELECT, INSERT, UPDATE, DELETE).

18. Explain the purpose of cartesian product in dbms.

- * Cartesian Product (Cross Join): Combines every row from one table with every row from another, resulting in all possible row combinations. Its primary purpose is as a foundational operation for various types of joins, though rarely used directly in queries without a WHERE clause.

19. Interpret the GROUP BY clause in SQL.

- * The GROUP BY clause is used with aggregate functions (e.g., COUNT, SUM, AVG) to group rows that have the same values in specified columns into summary rows. It allows calculations to be performed on each group.

20. State the MVD with an example.

- * MVD (Multivalued Dependency): Occurs when one attribute (or set) determines a set of values for another attribute, and this set is independent of other attributes.

- * Example: In (Employee, Skill, Project), if an employee has multiple skills and multiple projects independently, then Employee $\twoheadrightarrow$ Skill is an MVD.

21. Write down the main advantage and disadvantage of normalisation.

- * Advantage: Reduces data redundancy and improves data integrity by minimizing update/insert/delete anomalies.

- * Disadvantage: Can lead to increased query complexity due to more table joins, potentially affecting read performance for certain queries.

22. Explain partial and transitive dependency with a suitable example.

- * Partial Dependency: A non-key attribute depends on only part of a composite primary key (e.g., ProductID determines ProductName in an (OrderID, ProductID) key). Violates 2NF.

- * Transitive Dependency: A non-key attribute depends on another non-key attribute, which in turn depends on the primary key (e.g., EmployeeID $\rightarrow$ DeptName $\rightarrow$ DeptLocation). Violates 3NF.

23. Express view in DBMS. How does it differ from a table.

- * View: A virtual table based on a SQL query, storing only the query definition, not actual data.

- * Difference: Views don't store data physically, while tables do. Views are derived, tables are base units.

24. Explain trivial functional dependency with an example.

- * Trivial Functional Dependency: A functional dependency $A \rightarrow B$ where B is a subset of A. It is always true by definition and provides no new information about data relationships.

- * Example: {StudentID, CourseID} $\rightarrow$ StudentID.

25. Write the SQL statements to create a VIEW.

```
CREATE VIEW Active_Customers AS
SELECT CustomerID, CustomerName, Email
FROM Customers
WHERE Status = 'Active';
```

26. Differentiate between shared and exclusive lock in locking protocol.

- * Shared Lock (S-Lock): Allows multiple transactions to read data concurrently (for reading only).

- * Exclusive Lock (X-Lock): Allows only one transaction to access/modify data at a time, blocking all other locks (for writing).

27. What is “commit” and “rollback”?

- * COMMIT: Permanently saves all changes made during a database transaction, making them visible to other users and durable.

- * ROLLBACK: Undoes all changes made during a transaction, reverting the database to its state before the transaction began, typically used on error or cancellation.